

DROP Drop both $W_{tree} \times VPD$ and $W_{bg} \times VPD$
Remove RQ2; report average daytime effects. Ruling is irreversible after coefficient access.

Prospective decision rule

Before these diagnostics were computed, V6.2-D002 required both Gate A cities to have at least 30 complete passes in the 25° candidate design, VPD VIF ≤ 5 , maximum condition index ≤ 30 , leave-one-pass-out nonlinear VPD concurvity $R^2 \leq 0.80$, residual VPD SD ≥ 0.25 kPa, and continuously supported VPD width ≥ 0.50 kPa. Missing required geometry fails closed. Pairwise correlations are descriptive only.

Binding criterion	Threshold	Phoenix	Los Angeles
Complete full-design passes	≥ 30	4 / 44	1 / 48
VPD VIF	≤ 5	NE (85.05*)	NE (72.57*)
Maximum condition index	≤ 30	NE (37.09*)	NE (47.80*)
Nonlinear concurvity R^2	≤ 0.80	NE (0.966*)	NE (0.987*)
Residual VPD SD (kPa)	≥ 0.25	NE (0.308*)	NE (0.100*)
Common VPD width (kPa)	≥ 0.50	2.522	0.000

* Optimistic reduced-design value after omitting unavailable view and relative azimuth; nonbinding.

Why the branch fails

Neither city has a complete set of the required view-azimuth and relative sun–sensor azimuth controls (4/44 Phoenix; 1/48 Los Angeles), so the full design cannot be estimated. Even the favorable reduced check exceeds the VIF, condition-index, and nonlinear-concurvity ceilings in both cities. Los Angeles also fails the residual-SD and continuous-support floors.

Variation that would identify the interaction

The VPD interaction would be identified by within-city, cross-pass variation in exact-time VPD that remains after conditioning on exact-time air temperature and actual vapour pressure/dewpoint, local solar time, solar zenith and azimuth, view zenith, relative sun–sensor azimuth, and day of year—and only within a VPD range continuously shared across those time, solar-geometry, viewing-geometry, and seasonal conditions. That independent variation is not demonstrated here.

Supporting numbers

- Historical nonthermal 25° support: 44 Phoenix passes and 48 Los Angeles passes; 15° sensitivity: 19 and 27.
- VPD–air-temperature Pearson r : 0.948 Phoenix and 0.953 Los Angeles. VPD–day-of-year r : -0.039 and 0.495. Correlations alone do not determine the ruling.
- Reduced nonlinear VPD predictability is 0.966 and 0.987 out of sample; unexplained SD is 0.308 and 0.100 kPa.
- Common VPD support is 3.504–6.025 kPa in Phoenix (width 2.522); Los Angeles has no intersection (bounds cross at 1.653 versus 1.634 kPa; width 0).

Decision and downstream consequence

DROP the demand branch now. Both candidate interactions are removed together; RQ2 does not survive; high-demand p75 contrasts will not be reported; and later Stage 2 work is restricted to average daytime effects. The branch is not retained for publication appeal and cannot be restored after new coefficients are viewed.

Scope and traceability

Evidence is historical pass-level provenance, not the v6.2 study sample. Exact-time HRRR temperature and dewpoint were linearly interpolated from inherited bracketing-hour summaries; actual vapour pressure was computed from dewpoint. Solar coordinates were calculated at the frozen city centroids from local solar time and day of year. A geometry-only recovery read all 92 frozen passes from ECO_L1B_GEO.002 without opening LST: only 4/44 Phoenix and 1/48 Los Angeles passes had view and solar azimuth populated for every mapped view-valid cell. Partial arrays were not imputed or promoted to complete. Controlling sources: Deliverables Schedule p.3; Research Proposal pp.15–18; Working Guide pp.2–4, 7, 11–13. All input and output hashes are recorded in the package.

Status: DROP | V6.2-D002 rule | geometry recovery reassessment | D003 unchanged

Generated 2026-09-01 (Asia/Seoul). Historical evidence only; no thermal outcome access.